MoveOn Phone Cover Downloader & Translation Pipeline

Specification Version: 2.1 | **Project:** inventory/coverstore | **Status:** Production

1. System Architecture & Scope

Automates single-product asset acquisition from **moveon.global** with zero out-of-scope crawling. Features automatic material identification, interactive phone model selection, standardized SKU generation, and **in-place image translation** directly replacing original non-English pictures.

2. Custom SKU Generation Standard

All variation cover images are named using the standardized format:
[model abbreviation]-[material abbreviation]-[color abbreviation]-[print abbreviation]
Example: RN8-SIL-WT-S.jpg (Realme Neo8, Liquid Silicone, Antique White, Solid Color)

Component	Code Format	Representative Examples
Phone Model	2-6 Alphanumeric	`RN8 (Realme Neo8), IP15PM (iPhone 15 Pro Max), S24U (Galaxy S24 Ultra)`
Material	3 Letters	`SIL (Liquid Silicone), GLS (Glass), TPU (TPU), LTH (Leather), PC (Hard PC)`
Color	2 Letters	`WT (White), BK (Black), PR (Purple), GN (Green), BL (Blue), CL (Clear)`
Print / Finish	1 Letter	`S (Solid/Cube), P (Printed), G (Gradient), E (Electroplated/Mirror), C (Clear)`

3. Storage Hierarchy (Strict 2-Folder Structure)

Assets are organized into exactly two directories per model and material. Non-English text is translated in-place:

Directory Path	File Naming & Description
`\[model]\[material]\gallery\`	gallery_01.jpg ... Product showcase & gallery images. Any non-English text is translated in-place.
`\[model]\[material]\sku\`	[SKU].jpg (e.g. RN8-SIL-WT-S.jpg) ... Variant covers. Any non-English text is translated in-place.
`\plan\plan.pdf`	Technical specification document.
`\scripts\`	Python execution modules: downloader.py, sku_generator.py, moveon_client.py, translate_helper.py.